Optimal Route recommendation using GNN and user to user travel location recommendation

MDT6099 MASTERS' THESIS

Submitted by

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in partial fulfillment for the award of the degree of

M.Sc. Data Science

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April - 2024

DECLARATION

I hereby declare that the Project entitled **Optimal Route recommendation using GNN and user to user travel location recommendation** submitted by me to the Division of Mathematics, School of Advanced Sciences, Vellore Institute of Technology, Chennai Campus, 600 127 in partial fulfillment of the requirements of the award of the degree of **Master of Science in Data Science** is a bona-fide record of the work carried out by me under the supervision of **Dr. Sandip Dalui**. I further declare that the work reported in this project, has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma of this institute or of any other institute or University.

Place: Chennai

Date: 25/04/2024

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CERTIFICATE

This is to certify that the project entitled **Optimal Route recommendation using GNN and user to user travel location recommendation** is prepared and submitted by **Sagnik Chakravarty (Reg. No. 22MDT1009)** to Vellore Institute of Technology, Chennai Campus, in partial fulfillment of the requirement for the award of the degree of **Master of Science in Data Science** is a bona-fide record carried out under my guidance. The project fulfills the requirements as per the regulations of this University and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

Place: Chennai

Date: 25/04/2024

Signature of Guide

Dr. Sandip Dalui

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Dr. K. Muthunagai

Acknowledgement

With immense pleasure and a deep sense of gratitude, I wish to express my sincere thanks to my supervisor **Dr. Sandip Dalui**, Associate Professor, School of Advanced Sciences, Vellore Institute of Technology (VIT), Chennai for his motivation and continuous encouragement, this project would not have been successfully completed.

I am grateful to the Chancellor of VIT, **Dr. G.Viswanathan**, the Vice Presidents, **Mr. Sankar Viswanathan, Dr. Sekar Viswanathan, Dr. G V Selvam**, the Executive Director, **Dr. Sandhya Pentareddy**, the Assistant Vice President, **Ms. Kadhambari S. Viswanathan**, the Vice Chancellor **Dr. V. S. Kanchana Bhaaskaran**, the Pro Vice Chancellor of VIT Chennai, **Dr. T. Thyagarajan** and the Additional Registrar **Dr. P. K. Manoharan** for motivating me to carry out the project at Vellore Institute of Technology, Chennai.

I express my sincere thanks to **Dr.Mahalakshmi S**, Dean, School of Advanced Sciences, VIT, Chennai and **Dr. Muthunagai K.**, HOD, Mathematics and Computing, School of Advanced Sciences, VIT, Chennai for their support and encouragement.

I would like to express my sincere gratitude to **Mr Soven Singh**, at (Wondervogel) in Kolkata, for her guidance and support throughout this project.

I wish to extend my profound sense of gratitude to **my parents** for providing me the moral support and encouragement whenever required.

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Abstract

This thesis delves into optimizing travel routes to specific destinations, with a focus on Goa, a prominent tourist spot in India. Utilizing the OSMnx Python library for map data, I constructed a network graph of the area using the NetworkX package. By considering factors like traffic and road congestion, I identified the most efficient paths within this network. To achieve this, Graph Neural Networks (GNN) were employed, comparing their efficacy with traditional distance calculation methods. Furthermore, I employed a web crawler to gather user data from diverse tourism websites, including comments and descriptions. Subsequently, a neural network incorporating self-attention mechanisms was proposed to capture both short and long-term user preferences. This data was then used to train the model within the PyTorch deep learning framework. The model demonstrated notable accuracy, achieving 88% for Pre@10 and 83% for Rec@10, surpassing existing models in performance.

Keywords: *travel route optimization, OSMnx, NetworkX, graph neural networks, GNN, traffic analysis, road congestion, pathfinding, Goa, tourism, web crawling, user recommendation, personalized travel, India.*

Department of Mathematics, School of Advanced Sciences, Vellore Institute of Technology , Chennai

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Chapter 1

Introduction

1.0.1 Motivation for this Project

As our living standards continue to evolve, the appeal of tourism and recreational pursuits has experienced a significant upsurge. This surge is further fueled by the widespread adoption of the internet, with approximately 5.35 billion individuals globally and over 900 million people actively engaging online. This digital landscape empowers travelers to access a plethora of travel-related information directly from the web, reshaping how they plan their journeys.

In light of this digital revolution, there arises a need for a sophisticated system that not only provides travelers with the shortest routes to their desired destinations but also takes into account their specific points of interest. Such a system aims to optimize both travel costs and time for each individual consumer. Additionally, the steady rise in living standards and disposable incomes has fostered a greater inclination towards leisure activities.

While travelers can avail themselves of various tourist guides or online re-

sources to gather relevant information, the sheer volume of data available can pose a challenge in making well-rounded decisions about travel plans. Consequently, the latter part of our project endeavors to address this issue by offering personalized recommendations based on users' interests and their past visits to points of interest.

Through the analysis of similar users and their preferences, our system can effectively suggest comparable points of interest to travelers, enhancing their overall travel experience. This approach not only streamlines the planning process but also enhances the likelihood of discovering new and appealing destinations. In essence, by leveraging data-driven insights and user behavior patterns, our project seeks to elevate the efficiency and satisfaction levels of modern-day travelers

Development of route recommendation using graphs

To begin our journey towards finding the most efficient routes for travelers in Goa, India, we first delved into the rich pool of data available through the OSMNX library in Python. Given Goa's status as one of India's foremost tourist destinations, it served as an ideal backdrop for our exploration. Leveraging the OSMNX library, we meticulously extracted the intricate road network data that intricately weaves through Goa's picturesque landscapes.

With the aid of the networkx package, we meticulously constructed a comprehensive graph encapsulating the myriad pathways and junctions that adorn Goa's road network. This graph served as the backbone of our endeavor, laying the groundwork for our subsequent analyses and optimizations.

To streamline our process and focus our efforts, we distilled the vast network into a select set of top 50 nodes. These nodes, strategically chosen based on

various criteria, formed the nucleus around which our route optimization efforts revolved.

With our route framework in place, our attention turned towards understanding the temporal dynamics of traffic along these routes. Harnessing the power of OSMNX once again, we embarked on a quest to gather time-series traffic data for each segment of our selected routes. This endeavor aimed to provide us with invaluable insights into traffic patterns and congestion levels, crucial factors in determining the efficiency of travel routes.

As our journey unfolds, we will delve deeper into the realm of Artificial Neural Networks (ANN). These sophisticated algorithms, renowned for their ability to uncover intricate patterns and relationships within data, will serve as our guiding light in navigating the labyrinth of Goa's roadways. Through the adept utilization of ANN, we aspire to unearth the most optimal routes for travelers seeking to explore a set of randomly selected destinations within Goa's vibrant landscape. In doing so, we endeavor to not only streamline travel experiences but also to unlock new avenues for adventure and discovery amidst the captivating allure of Goa.

Significance of Travel Recommendation

Moving forward with our project, a pivotal milestone awaited us: crafting personalized travel recommendations for users. To achieve this, I harnessed the capabilities of the Scrappy package in Python, a versatile tool adept at web scraping. Through Scrappy, we embarked on a journey to extract a myriad of data points from the vast expanse of the internet, including but not limited to users' historical travel data, descriptions and reviews of destinations, captivating images capturing the essence of these locales, and more.

While numerous methodologies exist for generating travel recommendations, many grapple with inherent limitations such as compromised accuracy, timeliness, and adaptability, particularly in the context of intelligent robotic systems. Furthermore, certain models may encounter challenges stemming from long-term dependencies or sparse datasets, impeding their efficacy.

Addressing these challenges head-on, our approach leverages the power of big data analytics to overcome obstacles and pave the way for more robust and precise recommendations. Central to our strategy is the utilization of advanced techniques such as the Gaussian kernel factor node2vec model, alongside other cutting-edge technologies, to construct intricate embedded representations encompassing various dimensions such as user preferences, temporal dynamics, spatial characteristics, Point of Interest (POI) scores, access frequencies, and social relationships.

By harnessing the collective intelligence encoded within these embedded representations, we deploy them into the depths of a deep learning network for comprehensive analysis. This innovative approach not only mitigates the issue of low recommendation accuracy stemming from sparse data but also unlocks a realm of possibilities for enhancing user experiences and facilitating seamless exploration of diverse destinations.

As our journey unfolds, we stand poised at the forefront of a new era in personalized travel recommendations, fueled by the convergence of big data analytics, advanced machine learning techniques, and the boundless potential of the internet. Through our endeavors, we endeavor to redefine the landscape of travel planning, ushering in an era where each journey is meticulously tailored to the unique preferences and aspirations of every traveler.

Why Goa was chosen?

Selecting Goa as the nucleus for route recommendation within a deep learning model for India presents a myriad of unique advantages that set it apart as an ideal choice:

1. **Distinctive Tourist Appeal:** Goa's prominence as a tourist haven, celebrated for its sun-kissed beaches, vibrant culture, and historical landmarks, renders it an unparalleled magnet for travelers. Its allure extends across domestic and international spheres, ensuring a diverse influx of visitors with varying interests and inclinations.

2. **Abundant Diversity in Attractions:** From its idyllic beaches and verdant landscapes to its ancient forts and bustling markets, Goa boasts a kaleidoscope of attractions catering to an eclectic array of preferences. This diversity lends itself to the crafting of personalized travel itineraries tailored to the nuanced tastes of individual explorers.

3. **Rich Tapestry of Data**: Goa's stature as a tourism epicenter translates into a treasure trove of accessible data, spanning historical travel patterns, visitor feedback, geographical intricacies, and cultural nuances. This wealth of information serves as a pivotal resource for training deep learning models, facilitating the generation of precise and contextually relevant route recommendations.

4. **Complexity in Travel Planning**: Navigating Goa's myriad offerings and planning an itinerary can prove daunting, particularly for novices or those seeking off-the-beaten-path experiences. Factors such as transportation logistics, crowd dynamics, seasonal variations, and local festivities intricately shape the optimal route for exploration. Deep learning models adeptly unravel these complexities, discerning patterns and insights to curate bespoke recommendations that optimize

the travel experience.

5. **Discovery of Hidden Treasures**: Beneath the surface of Goa's well-trodden tourist trail lie a multitude of hidden gems and scenic enclaves awaiting discovery. Deep learning algorithms possess the capacity to unearth these lesser-known marvels by analyzing user preferences, historical data, and social interactions. In doing so, they enrich the traveler's journey by unveiling novel and enriching destinations.

6. **Cultural Enrichment and Heritage Appreciation:** Goa's cultural tapestry, woven through centuries of Portuguese influence and indigenous traditions, adds depth and authenticity to the travel narrative. Deep learning models, by integrating cultural insights into route recommendations, facilitate immersive encounters with local customs, culinary delights, and heritage sites. This fosters a profound appreciation for the region's historical legacy and cultural identity.

In essence, the selection of Goa as the focal point for route recommendation within a deep learning framework for India underscores its unparalleled blend of tourist allure, data richness, planning intricacies, scenic diversity, cultural significance, and potential for serendipitous discovery. Through the transformative power of deep learning, travelers are poised to embark on personalized odysseys imbued with authenticity, adventure, and unforgettable experiences.

This thesis aims to simplify gait mechanics and open the door to more useful clinical practice interventions by methodically gathering, analysing, and interpreting data.

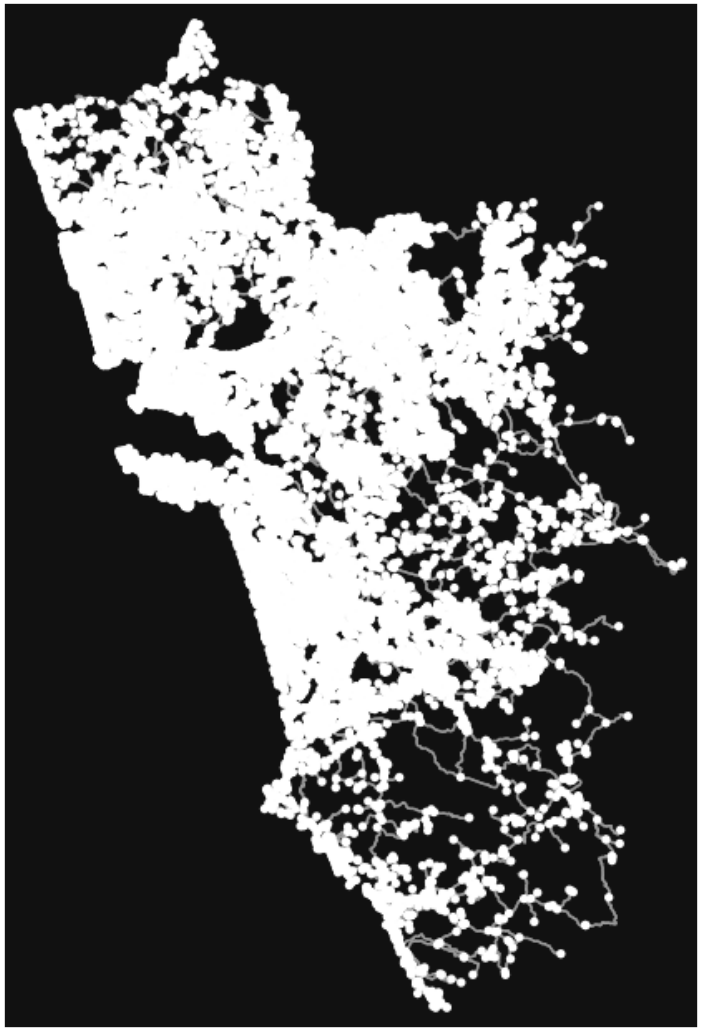

Figure 1.1: Goa Route Map

Project Objectives

The objectives of this project are:

1. **Extract Road Network Data**: Utilize the OSMnx package in Python to extract road network data for Goa, India, leveraging its popularity as a tourist

destination and the availability of rich data.

2. **Construct Graph Representation**: Employ the networkx package to construct a graph representation of the road network in Goa, providing a comprehensive framework for subsequent analyses.

3. **Identify Top Nodes**: Distill the vast road network into a select set of top 50 nodes, strategically chosen to serve as focal points for route optimization efforts.

4. **Collect Traffic Data**: Gather time-series traffic data for each segment of the selected routes in Goa using the OSMnx package, essential for understanding traffic patterns and congestion levels.

5. **Implement Deep Learning Model**: Develop a deep learning model, potentially utilizing techniques such as Artificial Neural Networks (ANN), to analyze the collected traffic data and identify the most optimal routes for visiting a specified number of random places.

6. **Scrape Internet Data**: Utilize the Scrappy package to scrape relevant travel data from the internet, including users' past travel data, descriptions and comments of places, images, and more.

7. **Overcome Limitations**: Address inherent limitations in existing methodologies for travel recommendation, such as limited accuracy, timeliness, adaptability, and challenges with long-term dependencies or sparse data.

8. **Utilize Big Data**: Leverage big data analytics to construct embedded representations of various dimensions, including user preferences, time, space, Point of Interest (POI) scores, access frequency, and social relationships.

9. **Deploy Deep Learning Network**: Deploy the deep learning network to analyze embedded representations and generate personalized travel recommendations tailored to individual user preferences and past travel experiences.

10. **Enhance Recommendation Accuracy**: Enhance recommendation accuracy by mitigating issues stemming from sparse data, thus providing travelers with more precise and contextually relevant route recommendations.

1.1 Literature Review

1.1.1 Deep Learning Boosts Personalized Travel Route Recommendations for Intelligent Service Robots in Tourist Sites : A Review

The paper [10] presents an innovative framework for personalized travel route recommendations, specifically tailored for intelligent service robots, within the context of a big data environment.

At its core, the model harnesses the power of deep learning algorithms to analyze vast datasets containing a wealth of information, including user preferences, historical travel behaviors, and various contextual factors such as time of day, geographical location, and social connections. By processing this rich trove of data, the model aims to generate highly personalized and contextually relevant travel route recommendations that cater to the unique needs and preferences of individual travelers.

Moreover, the utilization of deep learning techniques allows the model to un-

cover intricate patterns and relationships within the data, enabling it to discern subtle nuances and preferences that may elude traditional recommendation systems. This deep understanding of user behavior, coupled with the scalability and efficiency afforded by big data technologies, empowers the model to deliver recommendations with unprecedented accuracy and efficacy.

Furthermore, the paper emphasizes the importance of adaptability and scalability in the face of ever-evolving user preferences and dynamic travel environments. By operating within a big data framework, the model can continuously learn and adapt to changing circumstances, ensuring that its recommendations remain relevant and timely in an increasingly complex and dynamic travel landscape.

In summary, the proposed personalized travel route recommendation model represents a significant advancement in the field of intelligent service robotics, leveraging deep learning and big data technologies to enhance the travel experience for users. Its ability to deliver highly personalized and contextually relevant recommendations has the potential to revolutionize the way travelers plan and experience their journeys, ushering in a new era of intelligent and adaptive travel assistance.

1.1.2 Prediction Method using Point of Interest Recommendation

The paper titled [4] introduces a novel approach to point-of-interest (POI) recommendation by proposing a coarse-to-fine user preferences prediction method. The authors address the challenge of accurately predicting user preferences for POIs by developing a multi-stage prediction framework.

In the first stage, the model employs a coarse-grained strategy to predict user preferences based on high-level features such as user demographics and historical preferences. Subsequently, in the fine-grained stage, the model refines its predictions by incorporating more granular information, including user interactions and feedback with specific POIs.

The proposed method is designed to adapt to the dynamic nature of user preferences and provides personalized recommendations by iteratively refining its predictions. Experimental results demonstrate the effectiveness of the approach in improving recommendation accuracy compared to baseline methods.

Overall, the paper contributes to the field of POI recommendation by introducing a sophisticated prediction method that leverages both coarse and fine-grained features to provide personalized recommendations tailored to individual user preferences.

1.1.3 Graph Neural Network

The paper titled [16] offers a comprehensive overview of the methods and applications of Graph Neural Networks (GNNs). Here's a summary:

The authors provide an extensive examination of the evolving landscape of GNNs, beginning with an introduction to the fundamental concepts and motivations behind their development. They elucidate how GNNs have emerged as a powerful tool for learning representations from graph-structured data, offering a departure from traditional neural network architectures.

The review encompasses a detailed exploration of various GNN architectures and methodologies, including Graph Convolutional Networks (GCNs), Graph Attention Networks (GATs), GraphSAGE, and Graph Isomorphism Networks (GINs),

among others. Each architecture is dissected to reveal its underlying principles, strengths, and limitations, providing readers with a nuanced understanding of the diverse approaches to graph representation learning.

Moreover, the paper delves into the wide-ranging applications of GNNs across different domains, such as social networks, recommendation systems, bioinformatics, computer vision, and natural language processing. Through a series of case studies and examples, the authors highlight the versatility and effectiveness of GNNs in addressing various real-world challenges, ranging from community detection and link prediction to drug discovery and image recognition.

In addition to discussing existing methodologies and applications, the review also outlines current research trends, challenges, and future directions in the field of GNNs. This includes issues related to interpretability, scalability, and generalization, as well as emerging areas of research such as uncertainty estimation, multi-modal graph learning, and adversarial robustness.

In summary, ”Graph Neural Networks: A Review of Methods and Applications” serves as a comprehensive resource for researchers and practitioners seeking to gain insights into the theoretical foundations, practical implementations, and future prospects of GNNs. Through its meticulous examination of methods, applications, and challenges, the paper contributes to advancing the understanding and development of GNNs as a vital tool for graph representation learning and analysis.

1.1.4 Particle Swarm Optimization Technique

The paper titled [11] by Kennedy and Eberhart, presented at the ICNN’95 conference, introduces the concept of Particle Swarm Optimization (PSO). PSO is a

population-based optimization technique inspired by the social behavior of bird flocks and fish schools.

In PSO, a swarm of particles moves through a multidimensional search space, with each particle representing a potential solution to the optimization problem. The movement of particles is guided by their individual velocities and their best-known positions, as well as the best-known positions of the entire swarm. Through iterative updates to their velocities and positions, particles collectively converge towards the optimal solution.

PSO can be mathematically represented as follows:

$$v_{ij}^{t+1} = w \cdot v_{ij}^{t} + c_1 \cdot rand_1 \cdot (pbest_{ij} - x_{ij}^{t}) + c_2 \cdot rand_2 \cdot (gbest_j - x_{ij}^{t})$$
$$x_{ij}^{t+1} = x_{ij}^{t} + v_{ij}^{t+1}$$

Here, v_{ij}^{t} represents the velocity of particle i in dimension j at iteration t, x_{ij}^{t} represents the position of particle i in dimension j at iteration t, $pbest_{ij}$ represents the best-known position of particle i in dimension j, and $gbest_j$ represents the best-known position of the entire swarm in dimension j. w, c_1, and c_2 are constants representing the inertia weight, cognitive coefficient, and social coefficient, respectively, and $rand_1$ and $rand_2$ are random numbers between 0 and 1.

The PSO algorithm iteratively updates the velocities and positions of particles based on the above equations until a termination condition is met, such as reaching a maximum number of iterations or achieving a desired level of convergence.

Overall, the paper presents PSO as a novel optimization technique with applications in various domains, including artificial neural networks, optimization problems, and performance evaluation.

1.1.5 An Effective POI Recommendation for Heterogeneous Social media Network

The paper titled [14] presents a novel approach to point-of-interest (POI) recommendation tailored for heterogeneous social networks. The framework addresses the challenge of providing personalized recommendations by leveraging both social network information and user preferences.

The proposed framework integrates various factors, including user social interactions, geographical location, and POI characteristics, to generate contextually relevant recommendations. By modeling the heterogeneous relationships between users, POIs, and social interactions, the framework achieves enhanced recommendation accuracy and diversity.

Key components of the framework include graph-based representations of social networks and POIs, feature extraction methods, and machine learning models for recommendation generation. Through extensive experimentation on real-world datasets, the authors demonstrate the effectiveness of the proposed framework in improving recommendation quality compared to baseline methods.

Overall, the paper contributes to the field of POI recommendation by introducing a comprehensive framework that exploits the rich interconnectedness of heterogeneous social networks to deliver personalized and contextually relevant recommendations to users.

1.1.6 To Combine metadata For Better Recommendation accuracy

The paper titled [8] by D'Addio, Marinho, and Manzato, published in the journal Information Systems in 2019, proposes a method to improve recommendation accuracy by combining multiple metadata views.

The authors address the challenge of accurately recommending items to users by leveraging diverse metadata views associated with the items. By integrating various perspectives such as textual content, user behavior, and item attributes, the proposed method aims to capture a more comprehensive understanding of user preferences.

Key contributions of the paper include the development of a framework for combining different metadata views and the evaluation of its effectiveness in enhancing recommendation accuracy. Through empirical analysis on real-world datasets, the authors demonstrate the superiority of their approach compared to traditional recommendation methods.

Overall, the paper underscores the importance of leveraging diverse metadata views to enrich recommendation systems and highlights the potential of such approaches to improve the accuracy and relevance of recommendations for users.

1.1.7 Choose Scenic Routes using Route Choice Models

The paper [1] by Majid Alivand investigates the decision-making process of travelers when selecting scenic routes.

Alivand explores the factors influencing travelers' preferences for scenic routes by employing route choice models. These models analyze various variables such

as distance, travel time, road characteristics, and scenery attractiveness to understand how individuals make route decisions.

The study aims to uncover the underlying patterns and preferences of travelers when faced with multiple route options, particularly focusing on the significance of scenic beauty in their decision-making process. By applying route choice models, Alivand delves into the quantitative analysis of these preferences, shedding light on the factors that drive travelers to opt for scenic routes over more direct or efficient alternatives.

Through empirical analysis and data-driven insights, the paper contributes to a deeper understanding of the dynamics shaping travelers' route choices, particularly regarding the role of scenic beauty as a significant determinant in the decision-making process.

1.1.8 Neural Network Approach to Travel Route Recommendation

The paper titled[6] by Chen et al. proposes a novel neural network-based method for optimizing travel routes with multiple objectives.

The authors address the challenge of optimizing travel routes considering multiple factors such as time, cost, and convenience. They introduce a neural network approach to tackle this problem, leveraging its ability to learn complex patterns and relationships from data.

Key contributions of the paper include the development of a neural network model capable of incorporating multiple objectives into the route optimization process. By training the model on historical travel data, it learns to identify optimal routes that balance various objectives effectively.

Through empirical evaluations and comparisons with existing methods, Chen et al. demonstrate the effectiveness of their neural network approach in generating high-quality route recommendations that satisfy multiple objectives simultaneously.

Overall, the paper presents a promising approach to travel route optimization that harnesses the power of neural networks to address the complexities of real-world travel scenarios with multiple competing objectives.

1.1.9 Route Optimization using Reinforcement Learning

The paper titled [13] by Li et al. presents a novel approach for optimizing routes of tourist buses in densely populated urban areas through reinforcement learning.

The authors address the challenge of navigating tourist buses efficiently in crowded urban environments by leveraging reinforcement learning techniques. They propose a route optimization framework that utilizes reinforcement learning algorithms to iteratively learn and adapt optimal bus routes based on real-time traffic conditions, tourist attractions, and passenger demand.

Key contributions of the paper include the development of a reinforcement learning-based approach tailored specifically for tourist bus route optimization. By incorporating factors such as traffic congestion, tourist preferences, and operational constraints, the proposed framework enables the generation of dynamically adjusted routes that enhance both passenger satisfaction and operational efficiency.

Through empirical evaluations and case studies, Li et al. demonstrate the effectiveness of their approach in improving the performance of tourist bus operations in crowded urban areas. The results showcase significant improvements in

route efficiency, passenger waiting times, and overall travel experience.

Overall, the paper presents a promising solution to the challenge of route optimization for tourist buses in crowded urban areas, offering insights into the application of reinforcement learning techniques to enhance public transportation services in dynamic and complex environments..

1.1.10 Travel Recommendation based on Themic Interest

The paper titled [7] by X. Cheng introduces a novel travel route recommendation algorithm that considers both interest themes and distance matching.

Cheng addresses the challenge of providing personalized travel route recommendations by developing an algorithm that incorporates two key factors: interest themes and distance matching. The algorithm aims to recommend routes that align with the user's interests while optimizing travel distances.

Key contributions of the paper include the formulation of a recommendation algorithm that leverages interest themes extracted from user preferences and distance metrics to generate tailored travel routes. By combining these factors, the algorithm can provide more relevant and satisfying recommendations to users.

Through empirical evaluations and analysis, Cheng demonstrates the effectiveness of the proposed algorithm in generating high-quality travel route recommendations. The results show improved user satisfaction and relevance compared to traditional recommendation approaches.

Overall, the paper presents a promising approach to travel route recommendation that takes into account both user interests and distance considerations, offering potential enhancements to personalized travel planning experiences.

1.1.11 Recommendation based on Geo Tagged Images

The paper titled [13] by D. Lyu, L. Chen, Z. Xu, and S. Yu introduces a novel method for personalized travel location recommendation leveraging geo-tagged photos.

The authors address the challenge of providing personalized travel recommendations by developing a weighted multi-information constrained matrix factorization approach. This method integrates various types of user-related information, such as geo-tagged photos and user preferences, into the recommendation process.

Key contributions of the paper include the formulation of a matrix factorization model that incorporates multiple types of information and constraints to enhance the accuracy and relevance of travel location recommendations. By leveraging geo-tagged photos as a rich source of contextual information, the model can generate personalized recommendations tailored to individual user preferences.

Through empirical evaluations and comparisons with existing methods, Lyu et al. demonstrate the effectiveness of their approach in improving recommendation performance. The results show that the proposed method outperforms traditional matrix factorization approaches in terms of recommendation accuracy and user satisfaction.

Overall, the paper presents a promising solution to personalized travel location recommendation, offering insights into the integration of geo-tagged photo data and matrix factorization techniques to enhance the quality of recommendations for travelers.

1.1.12 Interurban Route Behaviour

The paper titled [2] by Ben-Akiva, Bergman, Daly, and Ramaswamy presents a comprehensive analysis of interurban route choice behavior.

The authors delve into the complexities of modeling the decision-making process of travelers when selecting routes between urban areas. They explore various factors influencing route choice behavior, including travel time, distance, cost, and personal preferences.

Key contributions of the paper include the development of mathematical models to capture the interplay of these factors and their impact on route choice. By applying advanced analytical techniques, the authors aim to enhance the understanding of traveler behavior and improve the accuracy of route choice predictions.

Through empirical analysis and case studies, Ben-Akiva et al. illustrate the applicability and effectiveness of their modeling approach in real-world transportation planning scenarios. The results highlight the importance of considering diverse factors and behavioral dynamics in route choice modeling to achieve more robust and reliable predictions.

Overall, the paper contributes to the field of transportation theory by offering insights into the intricacies of interurban route choice behavior and providing methodological advancements for modeling and analyzing this phenomenon.

1.1.13 Transition from destination prediction to route prediction

The paper titled [12] by Krumm, Gruen, and Delling explores the transition from destination prediction to route prediction in location-based services.

The authors investigate the evolution of location-based services and the increasing demand for route prediction capabilities. They highlight the limitations of traditional destination prediction methods and argue for the importance of accurately predicting routes to improve user experience and service efficiency.

Key contributions of the paper include the development of a framework for route prediction that considers not only destination preferences but also historical route choices and contextual factors. By integrating these factors into predictive models, the framework aims to generate more accurate and personalized route recommendations for users.

Through empirical evaluations and case studies, Krumm, Gruen, and Delling demonstrate the effectiveness of their approach in predicting routes with higher precision and relevance compared to traditional destination-based methods. The results underscore the importance of considering dynamic factors and user behavior patterns in route prediction algorithms.

Overall, the paper contributes to the advancement of location-based services by advocating for the adoption of route prediction techniques and offering insights into the design and implementation of effective route prediction frameworks.

1.1.14 Multi agent solution to address challenges associated with group

The paper titled [15]by Christian Villavicencio explores the concept of group recommender systems and proposes a multi-agent solution to address the challenges associated with recommending items to groups of users.

Villavicencio delves into the complexities of recommending items to groups, where preferences and interests of individual members may vary. The paper highlights the limitations of traditional recommender systems in catering to group dynamics and proposes a multi-agent approach as a solution.

Key contributions of the paper include the development of a multi-agent system where each agent represents a user in the group, capable of learning and adapting recommendations based on individual preferences and group dynamics. By leveraging collaborative filtering techniques and group modeling, the proposed approach aims to provide more accurate and personalized recommendations to groups of users.

Through theoretical analysis and simulation studies, Villavicencio demonstrates the effectiveness of the multi-agent solution in improving recommendation quality and satisfaction among group members. The results showcase the ability of the system to adapt to diverse group compositions and preferences, leading to enhanced recommendation performance.

Overall, the paper contributes to the advancement of group recommender systems by offering a novel multi-agent solution that addresses the complexities of recommending items to groups of users with varying preferences and dynamics.

1.1.15 Fuzzy greed algorithm to designing tourist trips with clustered POI's

The paper titled [9] by A. Expósito, S. Mancini, J. Brito, and J. A. Moreno presents a fuzzy Greedy Randomized Adaptive Search Procedure (GRASP) algorithm for designing tourist trips considering clustered Points of Interest (POIs).

The authors address the challenge of efficiently designing tourist trips that satisfy visitors' preferences while considering the spatial distribution and clustering of POIs. They propose a fuzzy GRASP algorithm that incorporates fuzzy logic to handle imprecise and uncertain information inherent in tourist trip planning.

Key contributions of the paper include the development of a fuzzy GRASP algorithm capable of generating high-quality tourist trip itineraries by effectively considering the spatial distribution and clustering of POIs. The algorithm optimizes trip routes by iteratively selecting and inserting POIs based on their fuzzy attractiveness values and spatial relationships.

Through empirical evaluations and case studies, Expósito et al. demonstrate the effectiveness of their approach in generating diverse and appealing tourist trip designs. The results highlight the ability of the fuzzy GRASP algorithm to adapt to varying user preferences and spatial distributions of POIs, leading to improved trip satisfaction and exploration experiences.

Overall, the paper contributes to the field of tourist trip planning by introducing a novel algorithmic approach that leverages fuzzy logic and GRASP techniques to design optimized trip itineraries considering clustered POIs.

1.1.16 Multi Criteria Framework for personalized POI's

The author [17] explores a multi-criteria framework for generating personalized recommendations of points-of-interest (POIs).

The authors address the challenge of providing personalized POI recommendations by developing the IMCRec framework, which considers multiple criteria in the recommendation process. These criteria include user preferences, spatial relevance, temporal relevance, and social influence.

Key contributions of the paper include the formulation of a comprehensive framework that integrates diverse factors to enhance the accuracy and relevance of POI recommendations. By leveraging a multi-criteria approach, IMCRec aims to cater to the diverse preferences and contexts of individual users.

Through empirical evaluations and comparisons with existing methods, Yan et al. demonstrate the effectiveness of IMCRec in generating personalized POI recommendations. The results highlight the ability of the framework to outperform traditional recommendation approaches by considering multiple dimensions of user preferences and contextual factors.

Overall, the paper contributes to the field of personalized recommendation systems by offering a multi-criteria framework that leverages various factors to generate more accurate and contextually relevant POI recommendations for users.

1.1.17 LSVP deep learning for Sparse data

The author [18] introduces LSVP, a deep neural direction learning model designed for point-of-interest (POI) recommendation using sparse check-in data.

The authors address the challenge of recommending POIs with limited check-

in data by leveraging visual information. LSVP utilizes deep neural networks to learn visual representations of POIs, enabling the model to make recommendations based on similarities between POI images and user preferences.

Key contributions of the paper include the development of LSVP, which integrates visual-based learning with deep neural networks to enhance POI recommendations. By learning from both check-in data and visual features of POIs, LSVP can provide more accurate and personalized recommendations, particularly in scenarios with sparse check-in data.

Through empirical evaluations, Yu et al. demonstrate the effectiveness of LSVP in improving recommendation quality compared to traditional methods. The results highlight LSVP's ability to leverage visual information to overcome data sparsity issues and generate more relevant recommendations for users.

Overall, the paper contributes to the advancement of POI recommendation systems by introducing LSVP, a novel model that combines visual-based learning with deep neural networks to address challenges associated with sparse check-in data.

1.1.18 Personalized POI's route recommendation for heterogenious tourist data

The paper titled [3] by C. Bin, T. Gu, Y. Sun, and L. Chang introduces a personalized point-of-interest (POI) route recommendation system leveraging heterogeneous tourism data and sequential pattern mining techniques.

The authors address the challenge of providing personalized recommendations for tourist routes by proposing a system that integrates diverse types of tourism data, including user preferences, POI attributes, and sequential patterns of tourist

behavior.

Key contributions of the paper include the development of a recommendation system that utilizes sequential pattern mining to extract temporal dependencies and preferences from historical tourist trajectories. By analyzing sequential patterns, the system can identify frequent sequences of POIs visited by users and generate personalized route recommendations based on these patterns.

Through empirical evaluations and comparisons with existing methods, Bin et al. demonstrate the effectiveness of their approach in generating personalized POI route recommendations. The results show that the proposed system outperforms traditional recommendation approaches by considering diverse types of tourism data and capturing temporal dependencies in tourist behavior.

Overall, the paper contributes to the field of personalized recommendation systems by offering a comprehensive approach that leverages heterogeneous tourism data and sequential pattern mining techniques to generate more accurate and relevant POI route recommendations for users.

1.1.19 Route location using GPS data

The paper titled [19] by Zheng, Huang, and Li explores tourist mobility patterns through GPS data analysis to predict their next destination.

The authors aim to enhance our understanding of tourist behaviors and preferences by analyzing GPS traces collected from tourists' mobile devices. They investigate the sequential patterns of tourist movements and propose a method to predict tourists' next destination based on their historical trajectories.

Key findings of the paper include insights into the spatial-temporal dynamics of tourist mobility, such as the tendency for tourists to visit nearby attractions

and the influence of time-of-day and day-of-week factors on destination choices. Additionally, the authors develop a prediction model based on machine learning techniques to forecast tourists' next destinations with reasonable accuracy.

Through empirical analysis and case studies, Zheng, Huang, and Li demonstrate the effectiveness of their approach in predicting tourists' future movements. The results highlight the potential applications of GPS data in tourism management and destination marketing, allowing for more targeted and personalized recommendations for tourists.

Overall, the paper contributes to our understanding of tourist mobility patterns and provides practical insights for tourism stakeholders to better cater to tourists' needs and preferences based on their real-time movements.

1.1.20 Tourist Recommendation using Hadoop technology

The paper titled [5] by Chen and Zhou presents the development and implementation of an intelligent system for recommending tourist routes using Hadoop technology.

The authors aim to address the challenge of providing personalized and efficient tourist route recommendations by leveraging big data processing techniques. They describe the design and implementation of a system that utilizes Hadoop, a distributed computing framework, to analyze large volumes of tourist data and generate tailored route recommendations.

Key aspects of the paper include the architecture of the intelligent system, which consists of data collection, preprocessing, analysis, and recommendation components. The system collects tourist data, preprocesses it to extract relevant features, analyzes the data using Hadoop-based algorithms, and generates person-

alized route recommendations based on user preferences and constraints.

Through empirical evaluations and case studies, Chen and Zhou demonstrate the effectiveness of their system in generating high-quality tourist route recommendations. The results highlight the scalability and efficiency of the Hadoop-based approach in processing large volumes of tourist data and providing timely and accurate recommendations to users.

Overall, the paper contributes to the field of tourist route recommendation systems by introducing an intelligent system that leverages Hadoop technology to analyze big data and generate personalized recommendations, thereby enhancing the tourist experience and satisfaction.

Chapter 2

Requirements

2.1 Software Used

- Python (including libraries such as NumPy, Pandas, Matplotlib, pytorch, osmnx, networkx, Microsoft Excel: Python was utilized extensively for data analysis, manipulation, and visualization. Libraries like NumPy and Pandas facilitated efficient handling of data structures, while Matplotlib was instrumental in creating various plots and visualizations. Additionally, osmnx was instrumental in gathering map data and networkx was instrumental in graph manipulation

2.2 Hardware Used

- Operating System: Windows 11: The choice of Windows 11 as the operating system provided compatibility with the required software tools and libraries.

- Processor: Intel Core i7: The Intel Core i7 processor facilitated efficient computation and data processing tasks, ensuring smooth operation of the software tools and algorithms.

- Hard Disk: Minimum 120 GB: A minimum of 120 GB of hard disk space was recommended to accommodate large datasets and software installations associated with the project.

- RAM: Minimum 16 GB: A minimum of 16 GB of RAM was necessary to support the processing and analysis of large datasets and to ensure smooth multitasking capabilities

- Graphic Card: Nvidia 3050Ti with 16 gb of ram was required.

2.3 Website Used

- WebPlotDigitizer: WebPlotDigitizer was utilized as a web-based tool for extracting numerical data from plots, graphs, and images. It provided a convenient way to digitize and extract data points from visual representations, facilitating further analysis and processing.

Chapter 3

Methodology

The process of route recommendation initiates with data collection facilitated by the OSMNx package within the Python environment. Following this, the approach entails identifying the top 50 frequently visited destinations within Goa, India. Subsequently, the focus shifts towards streamlining the graph to a more manageable scale, comprising solely these 50 destinations. A formulated algorithm is then employed, leveraging time series historical data encompassing routes from the preceding year (specifically, 2023-24), to condense the multitude of potential routes to approximately 82 feasible options.

Upon gathering tourists' input data, it undergoes analysis within the site prediction module. Within this module, predictions are generated to anticipate the subsequent site to be visited, utilizing artificial neural networks (ANNs). The route optimization module receives the results from the site prediction module together with comprehensive tourist data, system limitations, and user preferences. In the route optimization module, particle swarm optimization (PSO) is used as the basic algorithmic approach, and each of the inputs is carefully analyzed to find

the best path to the suggested site.Through this comprehensive process, the system adeptly tailors route recommendations to suit individual tourist preferences, enhancing the overall tourism experience.

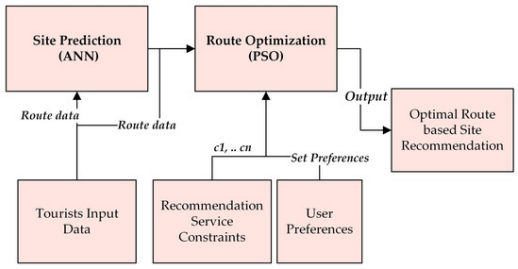

Figure 3.1: Route Recommendation

In the subsequent phase, data acquisition was conducted via web scraping techniques facilitated by the scrapy package within the Python programming environment. This sophisticated tool traverses various online platforms, systematically gathering pertinent tourist-related information dispersed across diverse websites. Subsequently, the collected data underwent processing through the node2vec model, an advanced algorithm designed to extract and analyze relevant tourist data alongside user-specific information. Through this meticulously orchestrated process, the system compiles a comprehensive dataset enriched with diverse insights, poised to fuel subsequent analytical and recommendation endeavors.

3.1 Route recommendation for Goa and Graph overview

Leveraging the versatile capabilities of the OSMNx package, I tapped into its vast repository of open-sourced map data spanning regions across the globe. This

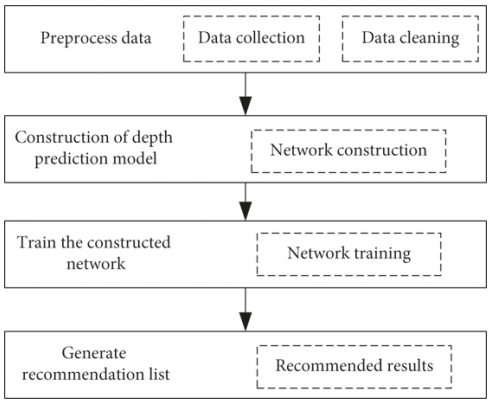

Figure 3.2: Travel Recommendation based on Web data

dynamic tool not only offers comprehensive map data but also provides the flexibility to tailor queries based on specific parameters such as traffic conditions, road congestion levels, and the density of traffic lights. Employing this robust platform, I embarked on a meticulous mapping endeavor covering the entirety of Goa, meticulously capturing longitudinal and latitudinal coordinates to meticulously delineate every road network within the state's bounds.

Intriguing insights emerged from this extensive mapping exercise. By scrutinizing the traffic flow patterns and analyzing the network's topology, I identified pivotal nodes characterized by significant inbound and outbound connections. These nodes, representing key points of interest, formed the crux of my analysis, with the top 50 most frequented nodes emerging as focal points for further investigation.

Moreover, an intriguing observation surfaced during this analysis: a discernible contrast in tourist attraction between the southern and northern regions of Goa. Evidently, the southern expanse boasted a more pronounced allure among tourists

compared to its northern counterpart. This revelation underscores the nuanced spatial dynamics at play within the state, offering valuable insights for subsequent tourism planning and development initiatives.

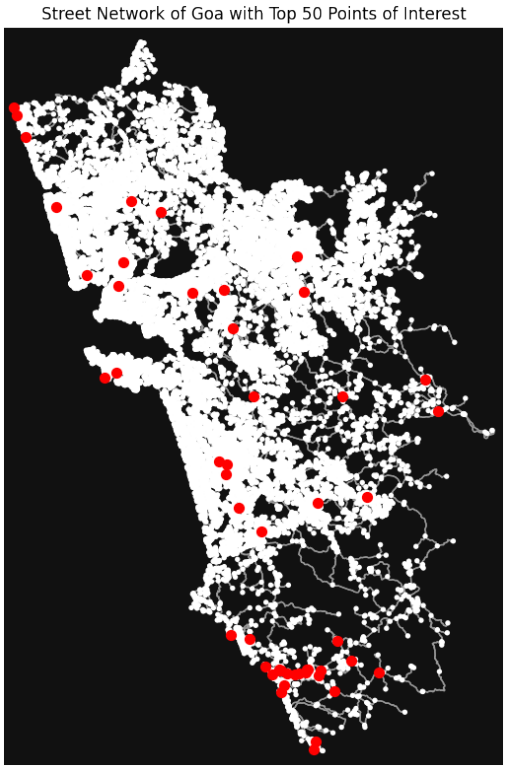

Figure 3.3: Most Popular tourist destination in Goa

3.1.1 Goa Route recommendation Model

Network Graph

- **Nodes and Edges**: Represent entities and relationships within the network.
- **Degree Distribution**: Reveals the frequency of different node degrees.

- **Clustering Coefficient**: Quantifies the tendency for nodes to cluster together.
- **Centrality Measures**: Assess the importance of nodes within the network.
- **Community Structure**: Identifies cohesive subsets of densely connected nodes.
- **Small-World Phenomenon**: Facilitates efficient communication with short average path lengths.
- **Scale-Free Property**: Exhibits a power-law distribution of node degrees, with a few highly connected nodes.

Artificial Neural Network

A computational model that draws inspiration from the architecture and operation of biological neural networks is known as an artificial neural network (ANN). It is made up of layers of interconnected nodes, also referred to as neurons or units. Until the final output is formed, each neuron processes incoming signals, computes, and then transfers the result to the layer of neurons below it. Mathematically, the output of a neuron in an ANN can be expressed as follows:

$$\text{Output} = f\left(\sum_{i=1}^{n} w_i x_i + b\right)$$

Where:

- x_i represents the input to the neuron.
- w_i denotes the corresponding weights associated with the inputs.

- b is the bias term.
- f is the activation function applied to the weighted sum of inputs and bias.

The activation function introduces non-linearity into the neural network, enabling it to learn complex patterns in the data. Two commonly used activation functions are the hyperbolic tangent (tanh) and softmax functions.

The hyperbolic tangent (tanh) activation function is defined as:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

For multi-class classification tasks, a neural network's output layer usually employs the softmax activation function. It calculates the probability distribution across several classes and verifies that the total of the output values equals 1.The softmax function is defined as:

$$\text{Softmax}(x_i) = \frac{e^{x_i}}{\sum_{j=1}^{K} e^{x_j}}$$

Where:

- x_i is the input to the softmax function for class i.
- K is the total number of classes.
- e denotes Euler's number, approximately equal to 2.71828.

These formulas describe the fundamental mathematical operations and activation functions used in artificial neural networks, enabling them to model complex relationships and solve various machine learning tasks.

Particle Swarm Optimization

Particle Swarm Optimization (PSO) is a population-based optimization technique inspired by the social behavior of birds flocking or fish schooling. In PSO, a population of candidate solutions, called particles, iteratively searches for the optimal solution within a predefined search space. Each particle adjusts its position based on its own experience and the experiences of its neighbors.

The position of each particle in PSO is represented as a vector $\vec{x} = (x_1, x_2, \ldots, x_n)$, where n is the dimensionality of the search space. The velocity of each particle is also represented as a vector $\vec{v} = (v_1, v_2, \ldots, v_n)$, indicating the direction and magnitude of movement.

At each iteration, particles update their velocities and positions according to the following equations:

$$\vec{v}_{ij}^{t+1} = w\vec{v}_{ij}^{t} + c_1 r_1(\vec{p}_{ij}^{t} - \vec{x}_{ij}^{t}) + c_2 r_2(\vec{g}_{ij}^{t} - \vec{x}_{ij}^{t})$$

$$\vec{x}_{ij}^{t+1} = \vec{x}_{ij}^{t} + \vec{v}_{ij}^{t+1}$$

Where:

- $\vec{v}_{ij}^{t}$ is the velocity of particle i in dimension j at iteration t.
- $\vec{x}_{ij}^{t}$ is the position of particle i in dimension j at iteration t.
- w is the inertia weight, controlling the impact of the previous velocity.
- c_1 and c_2 are the acceleration coefficients representing the cognitive and social components, respectively.
- r_1 and r_2 are random numbers between 0 and 1.

- $\vec{p}_{ij}^{t}$ is the best position encountered by particle i in dimension j up to iteration t.

- $\vec{g}_{ij}^{t}$ is the best position encountered by any particle in the neighborhood of particle i in dimension j up to iteration t.

These equations guide the particles towards promising regions of the search space, balancing exploration and exploitation. By iteratively updating their positions based on personal and neighborhood experiences, particles collectively converge towards the optimal solution.

Particle Swarm Optimization has been widely applied to various optimization problems, including function optimization, feature selection, and parameter tuning in machine learning algorithms, demonstrating its effectiveness in finding high-quality solutions in complex search spaces.

Creating the sub graph for the top 50 tourist destination

The edges are reduced by: The graph was reduced. First, the weights were assigned as follows:

$$\text{congestion_multiplier} = 1.5$$
$$\text{toll_weight} = 10$$
$$\text{signal_delay} = 30$$

To calculate the weight, the following steps were taken:

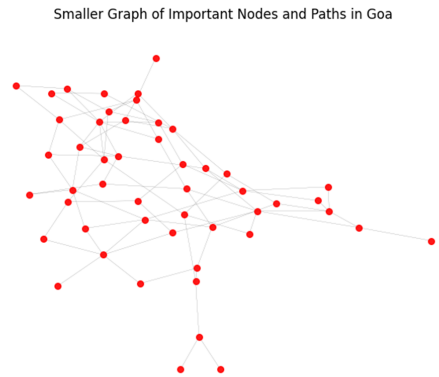

Figure 3.4: Top 50 nodes graph after excluding edges

$$\text{distance_km} = \frac{\text{distance}}{1000}$$

$$\text{free_flow_time} = \frac{\text{distance_km}}{\left(\frac{\text{free_flow_speed}}{3600}\right)}$$

$$\text{congestion_penalty} = \left(\frac{\text{travel_time}}{\text{free_flow_time}}\right) \times \text{congestion_multiplier}$$

$$\begin{aligned}\text{weight} = {} & \text{travel_time} + (\text{distance_km} \times \text{congestion_penalty}) \\ & + (\text{toll} \times \text{toll_weight}) + \left(\frac{\text{num_signals} \times \text{signal_delay}}{60}\right)\end{aligned}$$

We must generate a subgraph comprising the 50 tourist destinations along with their respective edges. This subgraph is constructed utilizing the Networkx package. The resulting graph is depicted below:

Tourist Site and Route recommendation model

The following delineates the process and setup for a site prediction module utilizing artificial neural networks (ANN) aimed at proposing forthcoming tourist destinations. Initially, tourist data is inputted into the system. After preprocessing, the data is divided into training and testing sets. The training data is then utilized by the learning module, employing ANNs to construct a trained model for the system. Once the learning phase concludes, the recommendation module receives the test data. At this stage, the module forecasts the next site to be suggested to the tourist.

The ANN-based site prediction module integrates various inputs, encompassing factors such as the day of the week, specific dates (with a focus on significant events), seasonal variations, historical route data, vehicle activity on the routes, and past tourist visitation figures along the routes. Consequently, the module generates the recommended tourist site as its output.

3.1.2 Data Processing and PSO pipeline

After identifying the optimal tourist attraction using the learning model, the subsequent step in the recommendation process involves determining the most effective route to reach the chosen tourist destination. The output from the ANN-based tourist site recommendation module is forwarded to the route optimization module, accompanied by supplementary inputs, to devise the most efficient route to the destination. The subsequent section outlines the overarching procedure for the route optimization module. Employed within this module is the PSO optimization algorithm, which factors in various elements including route data,

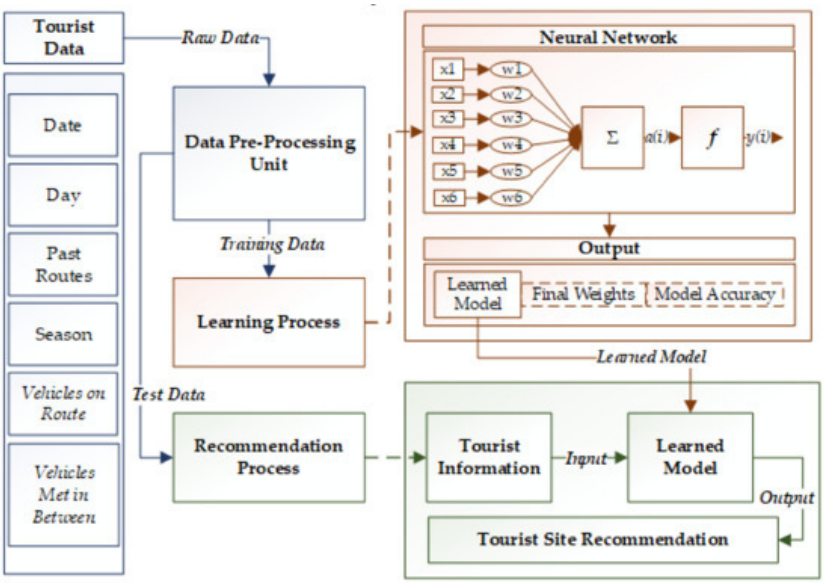

Figure 3.5: Tourist Site Recommendation using ANN

user preferences, the recommended site (obtained from the ANN module output), and a collection of service constraints. The route data comprises comprehensive details regarding all potential points connecting the tourist's current location to the recommended site. This encompasses a wide array of information such as weather conditions, traffic patterns, geographical features, and pertinent tourist-related data. User preferences entail inputs reflecting favored destinations and preferred routes, if provided. The recommended site signifies the destination selected for recommendation in the preceding module. Furthermore, the recommendation service constraints encompass a set of restrictions associated with specific tourist sites, such as visiting hours across different days and seasons, as well as the seasonal availability of certain attractions.

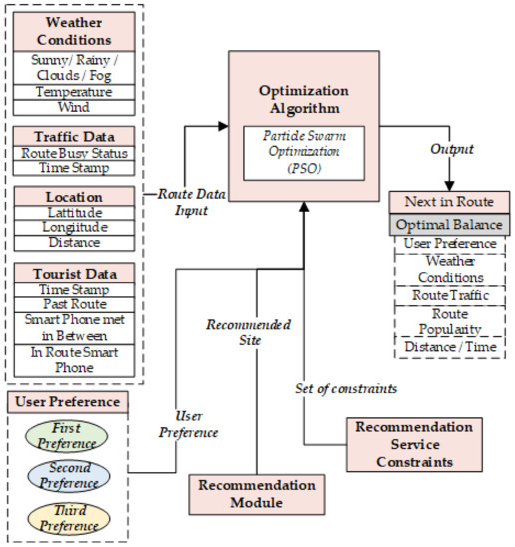

Figure 3.6: Route Optimization using Particle Swarm

Objective Function for Route Optimization

Let's initiate the analysis by examining two nodes, labeled as i and j, within a given route. Node i denotes the current location of the tourist, while node j signifies a potential destination. The term "distance" denotes the cumulative kilometers the tourist needs to traverse from node i to node j. "Road congestion" indicates the level of traffic density encountered between nodes i and j along the route. "Inclement weather conditions" refer to adverse weather occurrences experienced between nodes i and j during the journey. "User preference" reflects the individual's inclination towards the specific connection between nodes i and j. "Route popularity" quantifies the frequency at which the linkage between nodes i and j is utilized by other travelers. It's noteworthy that multiple connections may exist between nodes i and j, each characterized by distinct attributes such as distance, road congestion, weather conditions, user preference, and route popularity.

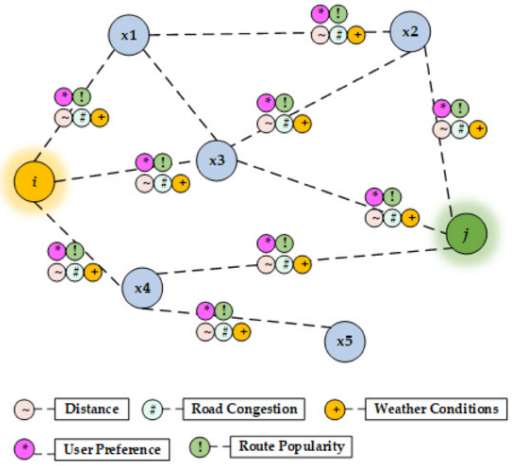

Figure 3.7: Graph Representation of Nodes

3.1.3 Route Optimization Using Particle Swarm Optimization

The flow that is shown demonstrates the PSO algorithm's operational sequence. The first stage of PSO is to create a population of particles, and then each particle's position and velocity are initialized. Next, each particle's fitness is evaluated according to its present position and velocity. Both the minimization and the maximization objective functions are used in the fitness evaluation. After this assessment, the best fitness of each particle (called pbest) is compared to its current fitness. The iteration continues if the particle's current fitness is greater than its pbest; if not, the pbest value is modified. The global best fitness (gbest) is compared with the updated pbest values. If a particle's pbest performs better than the gbest, the iteration proceeds and the gbest value is changed to match the particle's pbest. The particle's position and velocity are modified during each cycle in order to determine new fitness values.

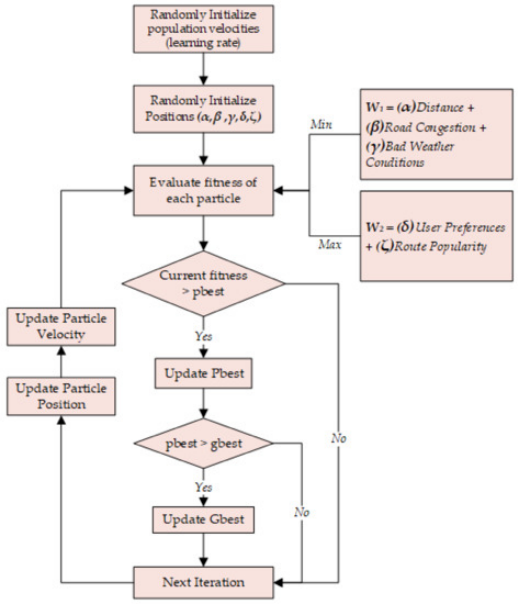

Figure 3.8: Route Optimization using PSO Flowchart

Our main objective is

$$w = min(w_1) + min(w_2) \tag{3.1}$$

Data Set amnd Preprocessing

The data collection and preprocessing step is illustrated below:

3.2 Personalized Travel Recommendation

Our subsequent goal entails the establishment of a personalized travel recommendation system. In the forthcoming sections, we will delve into intricate details

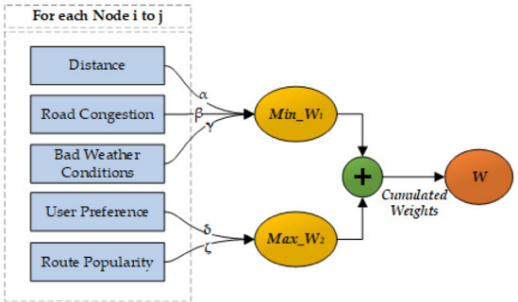

Figure 3.9: Objective Function for the route optimization

regarding the methodologies employed for network construction, training procedures, and data preprocessing steps. This comprehensive approach will encompass the implementation of robust strategies aimed at enhancing the effectiveness and accuracy of our travel recommendation system. By elucidating the intricacies of network building, training methodologies, and data preprocessing techniques, we aim to provide a thorough understanding of the underlying processes involved in the development of our personalized travel recommendation system. Through meticulous analysis and detailed explanations, we endeavor to equip readers with the knowledge and insights necessary to comprehend and replicate the proposed system effectively.

3.2.1 Data Collection for the personalized travel recommendation problem

The dataset collected consists primarily of two key components.

The first segment encompasses detailed information about users, including their gender, age, occupation, city of residence, and a history of their comments. Additionally, it includes textual feedback provided by users regarding their pre-

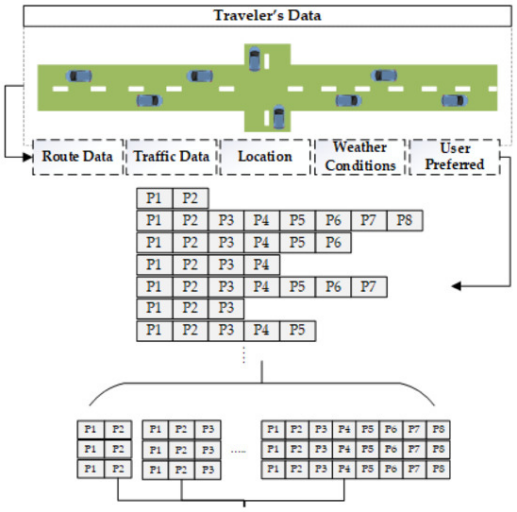

Figure 3.10: Data Collection and preprocessing

vious travel experiences, along with corresponding ratings. This section of the dataset serves as a crucial resource for extracting insights into user behavior, identifying preferences, and constructing comprehensive user profiles.

In the subsequent segment, the dataset contains essential details about various tourism service items, such as their names, locations, and categorizations. Moreover, it includes comment data associated with each service item, comprising textual reviews and accompanying ratings provided by users. This dataset facilitates the extraction of distinct attributes related to tourism services and supports the creation of characteristic models for such services, aiding in better understanding their features and quality.

To collect this extensive dataset, sophisticated web crawlers are employed to systematically explore relevant travel websites. Leveraging the Scrapy web crawler framework, which is built on Python, these crawlers exhibit exceptional flexibility and control, enabling efficient extraction of data from diverse online

sources. Notably, Scrapy streamlines the intricate process of web crawling, allowing researchers to focus more on the data extraction aspect without being burdened by technical complexities.

3.2.2 Data Cleaning

To ensure the accuracy of the recommendations provided, it's crucial to work with data that is both comprehensive and dependable. Therefore, the initial dataset obtained from web crawling must undergo thorough cleaning to remove any incomplete or irrelevant entries. The data cleaning process involves several steps:

1. Firstly, users with incomplete basic information are filtered out. Incomplete basic information refers to missing attributes necessary for analyzing the characteristics of tourism service items and users. Having complete basic information is essential for building reliable models.

2. Spam comments are identified and removed. By examining the scraped comment data, it becomes apparent that genuine comments typically contain meaningful content, whereas spam comments often consist of single-word responses or lack any discernible sentiment. Thus, comments lacking substantial content are flagged as spam and discarded as they do not contribute meaningfully to model creation.

3. Special symbols within the comments are eliminated to filter out unnecessary noise. These symbols add little value to the characterization of comment content. Once the data cleaning process is completed, a refined dataset containing reliable and coherent information is obtained, ready for further analysis and processing.

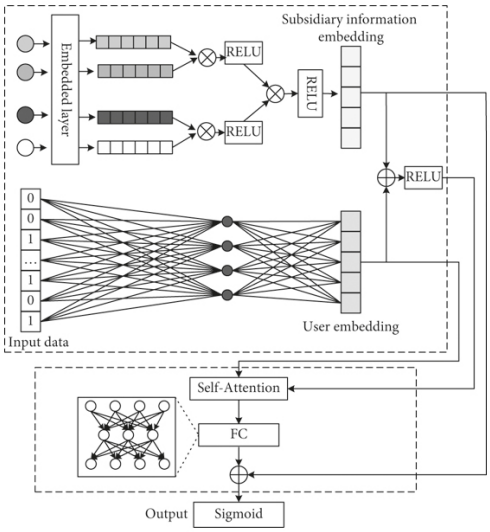

Figure 3.11: Network for Personalised Travel Recommendation

3.2.3 Building The Network

The proposed model consists of two primary elements: information embedding and information interaction. The model's structure is illustrated below.

First, multi-hot encoding is applied to the input data inside the information embedding module. In order to produce a possible representation matrix, a model for the user's point-of-interest (POI) check-in sequence is then built. Additionally, a Gaussian kernel function is used in the auxiliary information extraction phase to extract geographic position data associated with POIs, and softmax is used to normalize the extracted data.

Moving on to the information interaction module, self-attention mechanisms enable deeper data interaction, which enables the model to identify both short- and long-term user preferences. The final forecast is then generated by fusing the data through three bottleneck layers.

3.2.4 Embedding for User Representation

Let's consider the user representation embedding, denoted as $\mathbf{u}$, which acts as a latent vector capturing specific user preferences and characteristics. This embedding is derived from diverse user attributes represented by $\mathbf{X}_u$.

The user representation embedding $\mathbf{u}$ is computed using a neural network with parameters $\mathbf{W}_u$ and $\mathbf{b}_u$ according to the following equation:

$$\mathbf{u} = f(\mathbf{X}_u \cdot \mathbf{W}_u + \mathbf{b}_u)$$

Here:

- $\mathbf{X}_u$ represents the feature vector associated with the user, encompassing demographic data, past travel history, preferences, etc.
- $\mathbf{W}_u$ denotes the weight matrix.
- $\mathbf{b}_u$ signifies the bias vector.
- f denotes the activation function, such as ReLU, sigmoid, or tanh, introducing non-linearity into the user representation embedding.
- $\mathbf{u}$ emerges as the resultant user representation embedding vector, encapsulating the user's preferences and characteristics within a latent space.

This embedding plays a crucial role in capturing the user's nuanced preferences and traits, facilitating various travel recommendation tasks.

3.2.5 Spatio Temporal Embedding

Let's denote the spatio-temporal embedding as $\mathbf{v}$, which encapsulates the geographical and temporal characteristics of a point-of-interest (POI). This embedding is derived using a Gaussian kernel function to capture the spatial and temporal dynamics.

The spatio-temporal embedding $\mathbf{v}$ can be computed as follows:

$$\mathbf{v} = \sum_{i=1}^{N} \exp\left(-\frac{(x_i - x_{\text{user}})^2}{2\sigma_x^2} - \frac{(y_i - y_{\text{user}})^2}{2\sigma_y^2} - \frac{(t_i - t_{\text{user}})^2}{2\sigma_t^2}\right) \cdot \mathbf{e}_i$$

Here:

- (x_i, y_i, t_i) represents the spatial and temporal coordinates of the ith POI.
- $(x_{\text{user}}, y_{\text{user}}, t_{\text{user}})$ signifies the spatial and temporal coordinates of the user's current location and time.
- σ_x, σ_y, and σ_t are the bandwidth parameters controlling the spread of the Gaussian kernel along the spatial and temporal dimensions.
- $\mathbf{e}_i$ denotes the embedding vector associated with the ith POI.

This computation effectively captures the influence of nearby POIs on the user's current location and time, facilitating personalized travel recommendations based on spatio-temporal preferences.

Chapter 4

Results and Discussions

4.1 Optimal Route

In this section, we delve into analyzing the outcomes generated by our proposed route recommendation system, integrating Particle Swarm Optimization (PSO) and Artificial Neural Networks (ANNs). Initially, we present the preliminary assessment of the recommendation module's accuracy employing ANNs. Subsequently, we elucidate the results obtained from route optimization concerning the recommendation of tourist sites. Furthermore, we conduct a comparative analysis between PSO and Genetic Algorithm (GA) in terms of optimization efficacy.

Our dataset is divided into two subsets: a training set and a testing set, for the purpose of prediction-based recommendations using ANNs. 25% of the dataset is set aside for testing and 75% is allocated to the training set. There are six inputs, six hidden layers, and one output in our ANN implementation.

We use a total of 17 particles in the search space for the PSO implementation. The combination array for parameters $\alpha, \beta, \gamma, \delta, and \tau$ is included in the array that

represents the position of the search space in PSO particles.

4.1.1 ANN's prediction based on route size

We explore the site prediction performance results with respect to visitor data in this subsection. As previously indicated, our data is segmented based on route size during preprocessing.

As previously explained, our dataset includes multiple tourist trips throughout a whole year. Every trip can visit several places in a single day; the number of tourist sites seen in a day is known as the route size, and it can be anything from two to eight. With x tourist sites visited so far and other required input factors entered into our learning-based prediction model, the system should be able to forecast the next likely tourist site.

The accuracy of the predictions for different route sizes is shown above. The accuracy is approximately 68% for a route size of two and over 99% for a route size of eight. While larger input route sizes provide more data for the training phase and lower mistake rates, smaller route sizes limit the system's learning capacity.

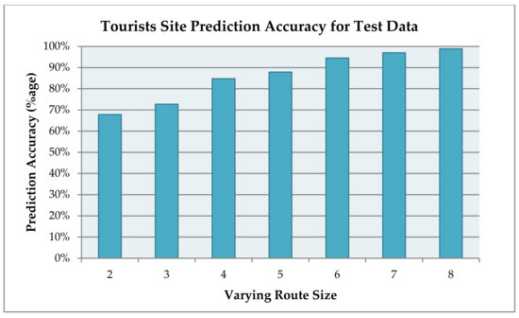

Figure 4.1: Accuracy for Test data with varying route size

The system remains reliable even with larger route sizes. Despite being limited to data for route sizes up to eight, we've generated additional test data for larger route sizes to ensure the system's adaptability. This adaptation enables the system to effectively handle larger route sizes while maintaining its efficiency.

Additionally, we evaluate and contrast the prediction performance of other methods, such as ANN, SVM, RF, and NB. The basis for these comparisons is the average forecast accuracy obtained from several test data iterations with six to eight route sizes. With the best prediction accuracy, the results unequivocally show that ANN is the best option for the given issue scenarios.The seasonal and festival-based learning prediction.

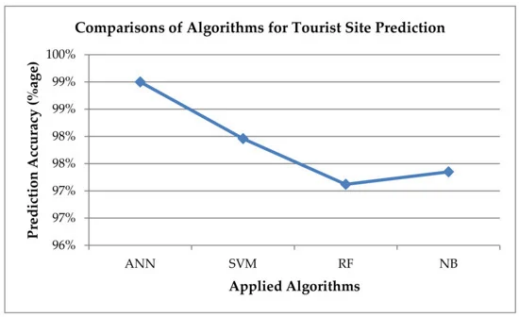

Figure 4.2: Route Prediction comparison against other ML algorithm

4.1.2 The Selected Sites for Recommendation

The chosen destinations for recommendation are cross-referenced with traveler feedback extracted from Goibibo and TripAdvisor. Goibibo is an Indian online platform widely utilized within India and offers a more extensive collection of records. The alignment of recommended destinations with traveler ratings from Goibibo and TripAdvisor is illustrated. The ratings range from 0 to 1 (Terrible), 1

to 2 (Poor), 2 to 3 (Average), 3 to 4 (Good), and 4 to 5 (Excellent). It's noticeable that a majority of the recommended destinations fall within the categories of good and excellent traveler ratings.

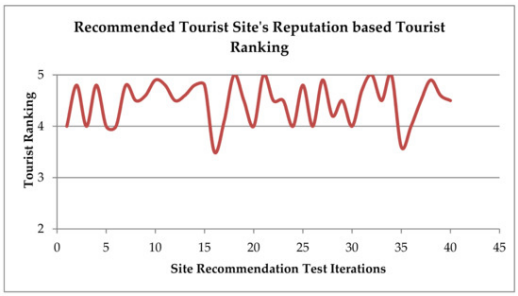

Figure 4.3: Tourist Site reputation based on ranking

4.2 Results for Personalised Travel Recommendation

The experimental data utilized originates from a tourism-oriented website, with Table:4.4 showcasing an illustration of the basic information pertaining to its users. Given the nature of tourist attractions, numerous comments are typically associated with each scenic spot. Likewise, individual users often provide multiple comments, with each comment being linked to specific user and scenic spot identities (IDs) along with corresponding scores. Due to space constraints, the table presented here provides only a condensed representation (with comments displaying solely the textual content).

Table 4.1: Tourism Site Data

Entry ID	Name	Label	Commentary
i984527	Taj Mahal	Iconic Monument	...“The Taj Mahal’s beauty is beyond words. A symbol of eternal love!” ...
i673892	Jaipur	Pink City Heritage	...“Exploring Jaipur’s colorful markets and majestic forts was unforgettable!” ...
i751239	Kerala Backwaters	Serene Waterways	...“Cruising through the Kerala Backwaters was a tranquil experience.” ...

The suggested model operates using the Pytorch 1.2.0 framework, with a fixed dropout rate of 0.5.

Evaluating Indicator

Two primary metrics are employed to assess the effectiveness of various recommendation models, specifically accuracy and recall (these metrics are denoted by Pre@N and Rec@N, respectively), and their computation is as follows:

$$\text{Pre@N} = \frac{1}{\theta}\sum_{u=1}^{\theta}\frac{|\text{Top-N}_u \cap K_u|}{N}$$

$$\text{Rec@N} = \frac{1}{\theta}\sum_{u=1}^{\theta}\frac{|\text{Top-N}_u \cap K_u|}{|K_u|}$$

where:

- θ represents the number of users.

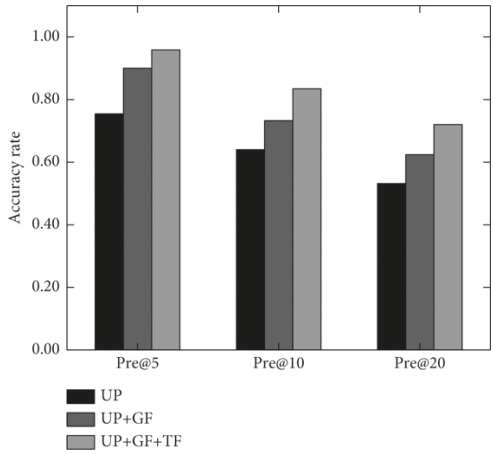

Figure 4.4: Model Characteristic check

- N represents the number of recommended points of interest (POIs).
- Top-N denotes the list of the first N POIs recommended by the model to the target user.
- K signifies the actual check-in list in the user test set, i.e., the set of POIs that the user has accessed in their historical records.

Influence of model Characteristics

Considering the impact of each feature on the recommendation outcomes, we progressively integrate user preferences (UP), geographic factors (GF), and thematic elements (TF) into our proposed model, conducting corresponding experiments. The comparative results for Pre@5, Pre@10, and Pre@20 are presented below.

4.2.1 Result for personalized route recommendation

For instance, considering Pre@10 and Rec@10, they stand at 88% and 83%, respectively, surpassing the values of other models, which are below 80%.The suggested model exhibits excellent overall recommendation performance by using the Gaussian kernel function to determine the pairwise distance between the points of interest (POIs) in the user check-in record, the node2vec model to extract the network structure features of the user's social connections, and the self-attention mechanism to capture user preferences.

Chapter 5

Conclusion and Future Work

Conclusion:

The dialogue has provided a comprehensive overview of travel route recommendation systems, highlighting their significance in delivering personalized travel experiences. Various algorithms, including Artificial Neural Networks (ANNs), Particle Swarm Optimization (PSO), and genetic algorithms, have been explored to optimize route recommendations. Crucially, data preprocessing steps such as cleaning, filtering, and feature extraction are pivotal in enhancing the accuracy and reliability of recommendation models. Evaluation metrics like accuracy and recall play a vital role in assessing the efficacy of these models. Moreover, integrating user preferences, geographical factors, and thematic elements has been identified as a means to augment recommendation performance.

Future Work:

Moving forward, there is ample scope for further research and innovation in this domain. Future endeavors could delve into integrating additional factors such as weather conditions, historical traffic data, and sentiment analysis to bolster recommendation systems' robustness and accuracy. Exploring advanced machine learning techniques like deep learning and reinforcement learning holds promise for enhancing route optimization and recommendation capabilities. Conducting user-centric studies and gathering feedback would offer insights into user preferences and needs, informing the refinement of recommendation algorithms. Additionally, efforts to enhance scalability and efficiency, particularly in handling large-scale datasets and real-time recommendation scenarios, are warranted. Lastly, incorporating context-awareness and temporal dynamics into recommendation models can pave the way for more personalized and timely recommendations, further enriching the travel experience for users.

Appendix A

Code Used

```
import osmnx as ox

# Define the place name for Goa, India
place_name = "Goa, India"

G = ox.graph_from_place(place_name, network_type='all')

# Plot the street network
ox.plot_graph(G)

# Basic graph statistics
basic_stats = ox.basic_stats(G)
print("Basic Graph Statistics:")

for key in basic_stats.keys():
```

```
  print(f'{key} : {basic_stats[key]}')

import matplotlib.pyplot as plt
import networkx as nx
# Compute degree centrality for each node
degree_centrality = nx.degree_centrality(G)

# Sort nodes by degree centrality and get the top 50
top_nodes = sorted(degree_centrality,
key=degree_centrality.get, reverse=True)[:50]

# Plot the street network
fig, ax = ox.plot_graph(G, show=False, close=False)

# Highlight the top 50 nodes with the highest degree centrality
for node in top_nodes:
    x, y = G.nodes[node]['x'], G.nodes[node]['y']
    ax.scatter(x, y, c='red', s=50, zorder=5)

plt.title('Street Network of Goa with Top 50 Points of Interest')
plt.tight_layout()
plt.show()

# Compute degree centrality for each node
degree_centrality = nx.degree_centrality(G)
```

```
# Sort nodes by degree centrality and get the top 50
important_nodes = sorted(degree_centrality,
key=degree_centrality.get, reverse=True)[:50]

# Create a new graph containing only the top 50 important nodes
important_graph = nx.Graph()

# Add important nodes to the graph
for node in important_nodes:
    important_graph.add_node(node)



for source in important_nodes:
    for target in important_nodes:
        if source != target:
            shortest_path = nx.shortest_path(G, source=source,
            target=target, weight='length')
            important_graph.add_edge(shortest_path[0],
            shortest_path[-1])
# Plot the smaller graph with important nodes and paths
pos = nx.spring_layout(important_graph)
nx.draw(important_graph, pos, node_color='red',
node_size=30, edge_color='black', width=0.1,
alpha=0.9, with_labels=False)
```

```
plt.title('Smaller Graph of Important Nodes and Paths in Goa')
plt.tight_layout()
plt.show()

import networkx as nx

# Define the origin (most popular destination)
origin = important_nodes[0]

# Define the destinations
# (top 10 most popular destinations)
destinations = important_nodes[1:11]

for destination in destinations:
    shortest_path = nx.shortest_path(G, source=origin,
    target=destination, weight='length')
    print(f"Optimal route from {origin} to
    {destination}: {shortest_path}")

import random
def assign_random_weights(G):
    for u, v in G.edges():
        weight = random.random()
        G[u][v]['weight'] = weight
assign_random_weights(important_graph)
```

```
important_graph = nx.convert_node_labels_to_integers(important_grap
first_label=0, ordering='default', label_attribute=None)
def remove_edges_below_threshold(graph, threshold):
    edges_to_remove = []
    for edge in list(graph.edges(data=True)):
        if edge[2]['weight'] < threshold:
            edges_to_remove.append(edge[:2])
    for edge in edges_to_remove:
      graph.remove_edge(*edge)

remove_edges_below_threshold(important_graph,0.93)
pos = nx.spring_layout(important_graph)
nx.draw(important_graph, pos, node_color='red',
node_size=30, edge_color='black', width=0.1, alpha=0.9, with_labels=
plt.title('Smaller Graph of Important Nodes and Paths in Goa')
plt.tight_layout()
plt.show()
```

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